Can you predict the likelihood of a customer defaulting on a loan based on their financial data ?

Financial institutions need to predict loan defaults to mitigate risk and optimise lending decisions. In Africa’s rapidly growing financial markets, with diverse customer demographics and dynamic economic conditions, accurately assessing default risk is more important than ever.

In this challenge, we want you to develop a robust, generalisable machine learning model to predict the likelihood of loan defaults for both existing customers and new applicants. Beyond accurate predictions, we encourage yous to innovate by incorporating unique factors relevant to each financial market.

The objective of this challenge is to develop a machine learning model that accurately predicts the probability of loan default.

The top 10 winners, in addition to submitting their solution, will need to design and submit a credit scoring function, using their model's outputs and probabilities. This step involves binning model outputs into risk categories and proposing a scalable credit risk score.

By accurately predicting loan defaults, your work will enable the client organisation to create a credit scoring solution to evaluate risk more effectively, improving decisionmaking, reducing financial losses associated with high-risk lending, and allowing for expansion into new financial markets.

The challenge provider is a private asset manager that operates in several financial markets across Africa.

Evaluation

The evaluation metric for this competition is F1 Score.

For every row in the dataset, submission files should contain 2 columns: ID and Target.

Where 1 indicates a loan default.

Your submission file should look like this (numbers to show format only):

```
ID              Target

ID_C7AV4GEJP9    1

ID_AFVZYGLXXY    0
```

About

You will work with loan and customer data from Kenya (train and test set) and Ghana (test set). This split emphasises the need for models that generalise well, i.e. perform well across different countries and financial contexts.

You will notice that a customer ID can appear in multiple rows of the dataset, and for a given customer, a loan ID can also appear with different lender IDs. For example, we can have the same customer ID on three rows with the same loan ID on each but a different lender ID. This observation means that the loan is funded by three separate lenders.

Additional data pulled from the Federal Reserve Economic Data (FRED) portal are provided to enrich model performance and offer insights into features that impact on credit scoring.

The economic indicators are:

- FP.CPI.TOTL.ZG: Inflation, consumer prices (annual %)
- PA.NUS.FCRF: Official exchange rate (LCU per US$, period average)
- FR.INR.RINR: Real interest rate (%)
- AG.LND.PRCP.MM: Average precipitation in depth (mm per year)
- FR.INR.DPST: Deposit interest rate (%)
- FP.INR.LEND: Lending interest rate (%)
- FR.INR.LNDP: Interest rate spread (lending rate minus deposit rate, %)
- EG.USE.COMM.FO.ZS: Fossil fuel energy consumption (% of total)
- SL.UEM.TOTL.ZS: Unemployment rate

The data downloaded from FRED is saved in a separate CSV in the data section.

File Description

SampleSubmission.csv

This file is an example of what your submission file should look like.

Test.csv

This is the dataset on which you need to apply your trained model for prediction.

Train.csv

This is the dataset on which to train your machine learning model. You are welcome to enrich the dataset with economic data sourced from the Federal Reserve Bank on the countries of interest.

economic_indicators.csv

External data pulled from the Federal Reserve Bank on the countries of interest.

Starter_Notebook_.ipynb

A starter notebook showing how to load data and run a basic ML model.

VariableDefinitions.txt
A description of each variable contained in Train/Test.csv